ELMEHDAOUI Ahmed zine el abidine

elmehdaoui.ahmed77@gmail.com
linkedin.com/in/ahmed-elmehdaoui-234182278 +213 541 60 40 09 — Tixeraine, Algiers, ALGERIA
github.com/ELMEHDAOUIAhmed Date of Birth: 31 Jan 2000
elmehdaouiahmed.github.io/AhmedZinPortfolio

Summary

Software engineer specializing in mobile development, backend and API development, AI-driven automation, and deployment. Strong experience in software engineering: requirements analysis, writing functional specifications, UML/BPMN modeling, and designing robust, secure software architectures. Experienced in leading technical teams, delivering containerized microservices platforms, and integrating AI models into production systems in agile environments.

Experience

- **Professional Services - AI R&D Engineer @CaptureDoc** *08/2025 – Present*
  Designing and implementing cognitive document capture and intelligent process automation solutions leveraging AI for document understanding and workflow optimization. Fine-tuning and deploying multimodal language models for automated document classification, data extraction, and content analysis. Developing AI-driven automation pipelines integrating NLP and computer vision to enhance operational efficiency and reduce manual processing time, orchestrated through Tungsten Total Agility (TTA).
- **Internship @Hb Technologies — Mobile Driving License (mDL)** *11/2024 – 07/2025*
  Design and development of a **mobile driver's license solution** compliant with **ISO 18013-5**. Includes two mobile apps (holder & verifier) and a platform for the **issuing authority**, built using a **microservices architecture**. Backend in **GoLang**, apps in **Flutter**, facial recognition in **Python**. Secured through **PKI infrastructure** and **Redis** for key storage.
  - Facilitated secure international usage of digital licenses through adherence to global interoperability standards.
  - Reduced risks of fraud by leveraging PKI encryption, facial recognition and electronic signatures.
  - Gained hands-on expertise in advanced security protocols, microservice design, and automation.

Projects

- **eKYC — CaptureDoc × Djezzy**
  Tech Lead of a **100% local** electronic Know-Your-Customer (eKYC) platform. Businesses purchase a set volume of identity verifications and integrate them directly into their apps or websites to verify their customers. All data is processed in-country, ensuring full **data sovereignty**.
  - Led the solution architecture and designed a **microservices-based backend**, containerized and deployed with **Docker**.
  - Integrated multiple **AI models** to automate identity verification.
  - Built the **license management** system, allowing customers to purchase and consume a defined quota of verifications.
  - Built the backend following **security-by-design** principles, validated by **penetration testing with no vulnerabilities reported**.
  - Owned delivery end to end, from architecture design to **DevOps** and production deployment.
- **Digibilan — CaptureDoc × Société Générale Algérie**
  Intelligent processing of financial statements containing dense tabular data, reducing processing time from 45 min–1 h to under 5 min.
  - Integrated **vision-language models (VLMs)** to extract and structure complex tabular data from financial statements.
  - Designed and built the **backend** orchestrating the extraction pipeline and exposing the results through APIs.
  - Significantly reduced manual data entry and processing effort for the bank's analysts.

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Technical Skills

Languages: Python, Dart, GoLang, Java, C, PHP, HTML, CSS, JavaScript
Frameworks & Libraries: Flutter, JavaFX, Laravel, RiverPod, Flask, GORM, Gin, PyTorch, Transformers, Huggingface, LangChain, OpenCV, Pandas, NumPy, NLTK, spaCy, FaceRecognition
DevOps & Deployment: Docker, containerized microservices deployment, application packaging & release, license management
Databases: Oracle, PostgreSQL, MongoDB, Redis
Modeling: UML 2.0, BPMN 2.0
Core Skills: Software design, microservices architecture, agile methodologies, RESTful API development, software architecture, secure backend development, advanced SQL / data warehousing
Soft Skills: Technical leadership, problem solving, analytical thinking, effective communication, time management
Tools: Git, Docker, Windev, Arduino, Figma

Education

- **Master's Degree in Software Engineering** *09/2023 – 07/2025*
  University of Science and Technology Houari Boumediène
- **Bachelor's Degree in Information Systems and Software Engineering** *09/2020 – 07/2023*
  University of Science and Technology Houari Boumediène
- **Scientific Baccalaureate – Experimental Sciences** *07/2019*
  Honours (Mention Bien)

Languages

English – Proficient
French – Good
Arabic – Native